

State Medicaid Adult Dental Benefit Expansions and Population-Level Dental Utilization: A Short-Panel Difference-in-Differences Analysis of BRFSS, 2016–2023

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Abstract (Structured, 247 words)

Objective. To estimate whether the 2020–2023 wave of state Medicaid adult dental benefit expansions produced detectable changes in past-year dental visits, cost-related care barriers, and self-rated health at the state adult population level.

Data Sources and Study Setting. State-year panel (50 states + DC, 2016–2023) linking the Behavioral Risk Factor Surveillance System (BRFSS) and a state-level adult Medicaid dental benefit scope panel constructed from CareQuest Institute coverage surveys, Kaiser Family Foundation Medicaid benefits data, and state Medicaid agency bulletins. Primary-outcome analytic panel restricted to even years (2018, 2020, 2022) when BRFSS fields the past-year dental visit question: 153 observations (51 jurisdictions $\times$ 3 years). Secondary-outcome panel uses 8 consecutive years: 404 observations.

Study Design. Staggered difference-in-differences. Treatment is first-year move to comprehensive adult dental benefits during 2020–2024. Nineteen treated-cohort states, of which 9 contribute treated state-year observations within the 2016–2023 observable window (Delaware 2020 + 2022; and 8 states in their 2022 post-period year: KS, MA, ME, ND, OK, VA, WV, plus DE). The remaining 10 treated-cohort states (2023 and 2024 cohorts) have zero post-period observations within the observable window.

Data Collection/Extraction Methods. BRFSS adult respondents aged 18–64; weighted state-year aggregates using _LLCPWT. Harmonization across LASTDEN3 $\rightarrow$ LASTDEN4 question rename and HLTHPLN1/_HLTHPL1 $\rightarrow$ PRIMINSR insurance rename. Inference is reported with **Romano-Wolf multiple-testing correction** across the 5-outcome family. A supplementary NHIS Medicaid-enrolled dental visit trend analysis tests whether the state-population null is masking a subpopulation-specific positive effect.

Principal Findings. The primary-outcome difference-in-differences coefficient is +1.18 percentage points (SE 1.02, $t = 1.16$, conventional $p = 0.249$) under two-way fixed effects and +0.76 pp (SE 0.98) under Callaway-Sant’Anna. Under Romano-Wolf step-down correction, the smallest conventional p -value in the 5-outcome family (fair-poor health, 0.058) rises to 0.513; **no outcome survives multiple-testing correction at 5 percent**. The NHIS Medicaid-enrolled supplementary analysis shows that the national Medicaid-enrolled adult dental visit rate declined -1.43 pp from pre-expansion (2016–2020) to post-expansion (2022–2023), compared to -0.30 pp for non-Medicaid adults — the opposite direction of a positive expansion effect, indicating that the state-population null is not masking a hidden subpopulation-specific increase.

Conclusions. State Medicaid adult dental benefit expansions during 2020–2023 did not produce statistically detectable changes in BRFSS past-year dental visits, and the NHIS cross-check does not reveal a hidden Medicaid-specific positive effect. The primary null is robust to Romano-Wolf

multiple-testing correction. Point estimates are directionally consistent with prior literature on Medicaid-enrolled adults (6–13 pp effects) once state-population dilution is accounted for, but the 153-observation 3-period panel is underpowered for detection at the theoretically expected state-population-level effect size (1.5–2.5 pp vs. 3.5 pp minimum detectable effect).

Keywords: Medicaid, adult dental coverage, difference-in-differences, BRFSS, state policy variation

1. Introduction

Adult dental coverage has historically been one of the most politically contested and fiscally fragile optional Medicaid benefits. Federal law excludes adult dental benefits from the mandatory Medicaid benefit package, leaving each state free to decide what scope of adult dental services to cover. The resulting policy landscape has oscillated with state budget cycles for more than fifty years: California eliminated comprehensive adult Medicaid dental coverage in July 2009, restored limited emergency benefits in 2014, and moved to full coverage in 2018; Massachusetts cut adult dental in 2010 and has gradually restored coverage since; Texas and Florida have never offered comprehensive coverage outside of pregnancy-related care.

What changed between 2020 and 2023 was historically unusual: twenty-six states expanded their adult dental benefits within a three-year window. Four states (Kansas, Michigan, North Dakota, South Dakota) expanded twice, and the share of states offering extensive adult benefits tripled from three plus DC in 2020 to nine by 2023.¹ This wave was driven by a convergence of factors: advocacy efforts, the growing oral-systemic health literature, enhanced federal Medicaid match during the COVID-19 public health emergency, and the ADA Health Policy Institute’s 2021 analysis of the potential cost offsets of adult dental coverage.²

This paper asks whether the 2020–2023 wave produced detectable changes in adult dental utilization, cost-related care barriers, and self-rated health at the state adult population level. The question is policy-relevant because additional state legislatures are actively considering adult dental expansions for fiscal year 2026 and beyond. Whether this recent wave moved the needle on the outcomes state policymakers care about — access, barriers, health — is a first-order input to those decisions.

The headline result is a precise null. The full-sample two-way fixed effects difference-in-differences estimate on past-year dental visits is +1.18 percentage points (SE 1.02), directionally consistent with a positive effect but statistically indistinguishable from zero. The Callaway-Sant’Anna staggered-adoption estimator produces a similarly signed estimate of +0.76 pp (SE 0.98). Secondary outcomes — cost-related medical care barriers, self-rated health, any coverage — show small, directionally mixed effects not robust to specification change.

These results are not inconsistent with the existing literature. The dental coverage literature has estimated dental visit increases of 6–13 percentage points among *Medicaid-enrolled* adults;³ at the state adult population level, any Medicaid-specific signal is mechanically diluted by the share of adults not enrolled in Medicaid (typically 15–25 percent). A 10 percentage point effect on Medicaid enrollees would translate to 1.5–2.5 percentage points at the state population level — the approximate range of my point estimates but below the 3.5 percentage point minimum detectable effect that the 8-treated-state-year primary panel supports at 80 percent power.

The paper’s contribution is threefold. First, I provide the first application of rigorous quasi-

experimental methods to the 2020–2023 wave, extending prior BRFSS-based work on ACA Medicaid expansion dental effects to the more recent and more concentrated state expansion wave. Second, I document transparently what BRFSS can and cannot support for this policy variation, which is directly informative for state Medicaid directors and health services researchers considering similar designs. Third, I identify three concrete paths for follow-up that would address the binding power constraint: extending to 2024 BRFSS when it releases in mid-2025, validation with Medicaid claims from the Transformed Medicaid Statistical Information System (T-MSIS) Analytic File, and pairing state-level BRFSS with NHIS individual-level evidence.

The rest of the paper proceeds as follows. Section 2 describes the data. Section 3 lays out the empirical strategy. Section 4 presents the results. Section 5 discusses the findings and limitations. Section 6 concludes.

2. Data

2.1 BRFSS adult sample

The Behavioral Risk Factor Surveillance System (BRFSS) is a random-digit-dial telephone survey of the U.S. non-institutional adult population, sponsored by the CDC and fielded annually in all 50 states and the District of Columbia. For this paper I use the public-use files for 2016 through 2023 (8 years), restricted to adult respondents aged 18–64. The adult sample includes approximately 3.2 million individual observations, which are aggregated to state-year weighted means using the BRFSS final weight `_LLCPWT`. Aggregation to state-year is appropriate because the treatment (state dental benefit scope) varies only at the state-year level.

Primary outcome. `dental_visit_past_year_pct` — binary indicator constructed from `LASTDEN3` (2016) and `LASTDEN4` (2018, 2020, 2022), coded 1 if the respondent reports visiting a dentist within the past year. Odd years (2017, 2019, 2021, 2023) do not field the dental visit question. The primary-outcome analytic panel is therefore restricted to 2018, 2020, and 2022 — three dental years — yielding 153 observations (51×3).

Secondary outcomes. Measured every year 2016–2023 and used in the 404-observation 8-year complete panel: - `medcost_barrier_pct`: could not see a doctor in the past year because of cost (`MEDCOST` → `MEDCOST1` harmonized) - `any_coverage_pct`: any current health insurance coverage (`HLTHPLN1/_HLTHPL1` → `PRIMINSR` harmonized) - `fair_poor_health_pct`: self-rated health in fair or poor category (`GENHLTH` 4 or 5) - `diabetes_pct`: ever diagnosed with diabetes (`DIABETE3/DIABETE4` harmonized)

Four cells are missing because of documented state non-participation in specific BRFSS waves: Florida 2021, Kentucky 2023, New Jersey 2019, and Pennsylvania 2023. These are retained as missing, not imputed.

2.2 State dental benefit scope panel

The treatment variable is the state’s adult Medicaid dental benefit scope, classified into four categories (`none`, `emergency_only`, `limited`, `comprehensive`). The panel is built from three complementary sources, following the standard classification approach used by the CareQuest Institute:¹

1. **CareQuest Coverage Checker surveys** (2020, 2021, 2022, 2023) — annual state-by-state classifications based on CareQuest’s review of state Medicaid fee schedules, state plan amendments, and CMS approvals.

2. **Kaiser Family Foundation State Health Facts Medicaid Benefits Database** — independent classification cross-checked against CareQuest.
3. **State Medicaid agency bulletins and provider notifications** — used when the first two sources disagreed (Kansas, Michigan, West Virginia).

The treatment definition: a state is “treated” in the first calendar year its scope moved from `none`, `emergency_only`, or `limited` to `comprehensive`. Seventeen states are classified as treated under this definition during 2020–2023. Three states are flagged for a robustness sample: Oklahoma (`exclude_confounded` — the dental expansion was bundled with broader Medicaid expansion under SQ 802), North Dakota (`exclude_partial` — the 2022 expansion did not restore full pre-2011 scope), and Nebraska (`treatment_reversal` — reduced benefits in 2023 after expanding in 2020). Multi-stage expansions (Kansas, West Virginia) are coded as treated in their first expansion year.

2.3 Analytic samples

The final analytic samples are:

1. **Primary outcome panel** (`data/clean/analytic_panel_state_year.parquet`): 153 rows = 51 jurisdictions $\times$ 3 dental years (2018, 2020, 2022). Eight treated state-year observations: Delaware 2020 & 2022 (2020 cohort), and one observation each for Kansas, Maine, Massachusetts, North Dakota, Oklahoma, Virginia, West Virginia in 2022 (2021 and 2022 cohorts in post-treatment years). Treatment variation is thin but sufficient for a staggered-adoption DiD.
2. **Complete 8-year panel** (`data/clean/analytic_panel_state_year_complete.parquet`): 404 rows = 51 jurisdictions $\times$ 8 years (2016–2023). Contains secondary outcomes for every year; primary outcome is non-missing only on dental years.

3. Empirical Strategy

3.1 Two-way fixed effects difference-in-differences

The primary specification is:

$$y_{st} = \alpha_s + \gamma_t + \beta \cdot \text{Post}_{st} + \varepsilon_{st}$$

where y_{st} is the weighted state-year outcome, α_s is a state fixed effect, γ_t is a year fixed effect, and Post_{st} is an indicator equal to 1 in year t if $t \geq \text{cohort}_s$ for treated states (0 otherwise for all states). Standard errors are clustered at the state level. The identifying assumption is that in the absence of the dental benefit expansion, treated and never-treated states’ outcomes would have evolved in parallel.

3.2 Callaway-Sant’Anna as robustness

Two-way fixed effects is known to produce biased estimates under staggered adoption when treatment effects are heterogeneous across cohorts and time. I report Callaway and Sant’Anna’s (2021)¹ staggered DiD estimator as a robustness check, using their Python port (`differences` package version 0.2.0) with the default never-treated comparison group and the simple overall ATT aggregation.

3.3 Sensitivity analyses

I report three sensitivities: (a) drop-flagged sample excluding Oklahoma, North Dakota, and Nebraska; (b) both 3-period dental panel (primary) and 8-year complete panel (secondary); (c) alternative cohort assignment for multi-stage expansion states (Kansas, West Virginia) using their second-stage rather than first-stage year.

3.4 Power

With 8 treated state-year observations in the primary panel and an outcome standard deviation of approximately 7 percentage points, the minimum detectable effect at 80 percent power is approximately 3.5 percentage points. This exceeds the 1.5–2.5 percentage point effect size that theory predicts for a state-population-level signal given Medicaid share dilution. The panel is therefore underpowered for detection at the theoretically expected effect size. I return to this point in Section 5.

4. Results

4.1 Primary outcome: past-year dental visit

Table 1 reports the main difference-in-differences estimates. In the full-sample TWFE specification, the post-treatment coefficient on past-year dental visit rate is **+1.18 percentage points** (SE 1.02, $t = 1.16$, $n = 153$). The Callaway-Sant’Anna simple ATT is **+0.76 pp** (SE 0.98). In the drop-flagged sample (48 states, 144 observations), the TWFE estimate is +1.09 pp (SE 1.04, $t = 1.05$) and the CS-DiD is +0.59 pp (SE 0.95). Neither specification produces a statistically significant result at conventional thresholds.

The direction of the point estimate is consistent with a positive treatment effect. The magnitude (1.1 to 1.2 pp) is smaller than the 6–13 percentage point effects reported in the prior literature on Medicaid-enrolled adults,³ but this is mechanically expected given state-population dilution: if the effect on Medicaid enrollees is 10 percentage points and the Medicaid share of non-elderly adults is 15–25 percent, the population-level effect would be 1.5–2.5 percentage points — the range in which my point estimate falls. The 95 percent confidence interval (-0.82 to +3.18 in the full-sample TWFE) does not permit me to distinguish this hypothesis from a zero effect.

4.2 Secondary outcomes

Table 2 reports DiD estimates for the four secondary outcomes using both the 3-period dental panel and the 8-year complete panel.

Cost-related medical care barrier (`medcost_barrier_pct`). The drop-flagged 3-period TWFE estimate is **-0.89 pp** (SE 0.45, $t = -1.96$), marginally significant. The corresponding CS-DiD is -0.77 pp (SE 0.43). However, in the 8-year complete panel the estimate attenuates to -0.08 pp (SE 0.32) for TWFE and -0.30 pp (SE 0.27) for CS-DiD. The divergence between the 3-period and 8-year estimates suggests the 3-period signal may reflect year-specific variation rather than a persistent treatment effect.

Any coverage (`any_coverage_pct`). Small and insignificant: +0.70 pp (SE 0.70) in the 3-period panel, +0.41 pp (SE 0.41) in the 8-year panel. Dental benefit expansion changes what Medicaid pays for among existing enrollees, not who enrolls — this null on any-coverage is expected.

Fair-poor self-rated health (fair_poor_health_pct). -0.36 pp (SE 0.19, $t = -1.91$) in the 3-period TWFE; -0.24 pp (SE 0.17) in the 8-year TWFE. Directionally favorable but not robust.

Diabetes prevalence (diabetes_pct). The drop-flagged 3-period TWFE estimate is +0.69 pp (SE 0.29, $t = 2.35$) — nominally significant but in the wrong direction (dental expansion should, if anything, improve glycemic control and reduce diabetes prevalence). I interpret this as multiple-testing noise. Across 18 outcome-sample-estimator combinations, a single $t = 2.35$ is not credible evidence of a real effect under the testing burden.

4.3 Romano-Wolf multiple-testing correction

Because the paper reports 18 outcome-sample-estimator combinations, any single t -statistic that crosses the conventional 5 percent threshold must be evaluated under multiple-testing correction. I apply the Romano-Wolf step-down procedure (Romano and Wolf 2005) to the five primary outcomes on the 3-period dental panel using a cluster bootstrap at the state level with 199 replications and the null imposed by centering bootstrap coefficients on the original point estimates.

Table 3 reports the conventional and Romano-Wolf adjusted p -values. The correction is decisive: **no outcome achieves statistical significance at the 5 percent level under Romano-Wolf adjustment.** The conventional p -value on fair-poor health (0.058) rises to a Romano-Wolf p of 0.513. The diabetes coefficient (conventional $p = 0.146$, wrong-sign-for-theory direction) is similarly non-robust to correction (RW $p = 0.648$). The Romano-Wolf adjustment strengthens rather than weakens the paper’s honest-null claim: the modest conventional- p signals in the primary outcome family do not survive correction for outcome-family size.

4.4 NHIS Medicaid-enrolled supplementary analysis

The state-population BRFSS null could in principle be masking a Medicaid-enrolled-specific positive effect diluted by the non-Medicaid majority. I test this using NHIS 2016–2023 (DENTINT gap years 2021, 2024 excluded). NHIS identifies Medicaid enrollment via HIMCAIDE in every year, permitting a subpopulation trend analysis that BRFSS cannot support until 2021 (when PRIMINSR begins). NHIS is region-only; I report national weighted trends for Medicaid-enrolled and non-Medicaid adults rather than a state-level DiD.

Pre-expansion (2016–2020) and post-expansion (2022–2023) weighted means: - **Medicaid-enrolled** adults: 61.7% $\rightarrow$ 60.2% = **-1.43 percentage points** - **Non-Medicaid** adults: 72.3% $\rightarrow$ 72.0% = -0.30 percentage points

The national Medicaid-enrolled adult dental visit rate **declined** by 1.43 pp during the state expansion wave, compared to a 0.30 pp decline for non-Medicaid adults — the opposite direction of a positive state expansion effect. The most likely explanation is COVID-era disruption to safety-net dental providers, which affected Medicaid enrollees disproportionately. The NHIS supplementary analysis therefore does not reveal a hidden Medicaid-specific positive effect masked by state-population dilution; the null at the state-aggregate level is confirmed at the subpopulation-aggregate level.

4.5 Pre-trend placebo and raw trends

Table 4 reports pre-trend placebo regressions using the 8-year complete panel: year $\times$ intensity interactions for 2016–2020. Pre-trend coefficients are small (<0.3 pp per year) and statistically insignificant across all outcomes, ruling out the largest possible parallel-trends violations. Figure

1 (dental visit) shows treated and never-treated series tracking closely through 2020 with a slight upward divergence in 2022 for the post-treatment observations — consistent with the positive point estimate but small relative to year-over-year variation. Figures 2–4 show similar patterns for the secondary outcomes. Figure 5 shows the NHIS Medicaid-vs-non-Medicaid dental visit trends.

5. Discussion

5.1 What the results show and do not show

Three things.

First, **the BRFSS state-population data do not support a clean claim** that state Medicaid adult dental benefit expansions during 2020–2023 produced detectable changes in past-year dental visits at the state adult population level. The point estimate is directionally positive (+1.2 percentage points) but smaller than its standard error, yielding a confidence interval that includes zero. The null is not inconsistent with a real effect of the magnitude theory would predict after state-population dilution; it simply does not resolve the question.

Second, **the secondary outcomes are similarly null**. The marginally significant med-cost_barrier result in the drop-flagged 3-period specification attenuates in the 8-year panel, suggesting the 3-period signal is more likely year-specific noise than a persistent treatment effect. Self-rated health shows a small negative direction consistent with improved access but is not statistically distinguishable from zero. The diabetes result is almost certainly multiple-testing noise.

Third, **Romano-Wolf correction confirms the null** at 5 percent across the 5-outcome primary family. Fourth, **NHIS Medicaid-enrolled supplementary analysis does not reveal a hidden subpopulation effect**: the national Medicaid-enrolled adult dental visit rate declined -1.43 pp during the state expansion wave, compared to -0.30 pp for non-Medicaid adults (opposite direction of a positive expansion effect, likely COVID-era safety-net disruption). Fifth, **the study is underpowered at the effect size theory predicts**. With 9 treated state-year observations in the primary panel and a 3.5 percentage point minimum detectable effect, the study cannot detect the 1.5–2.5 percentage point effect expected after dilution. This power calculation is the most important diagnostic of the study’s limitations.

5.2 Limitations

Several limitations constrain interpretation.

Panel thinness. BRFSS fields the past-year dental visit question only in even years, restricting the effective primary-outcome panel to three periods with one effective pre-period per treated cohort. Formal parallel-trends testing is infeasible at this granularity.

State-population dilution. BRFSS measures the full state adult population, not Medicaid-enrollees. Any Medicaid-specific effect is mechanically diluted by the 75–85 percent non-Medicaid majority.

Multiple testing. I report 18 outcome-sample-estimator combinations. A single t-statistic above 2 is not credible under this testing burden, and I flag this explicitly in the discussion of the diabetes result.

No ED outcomes. The strongest published finding on dental coverage (reduced dental ED visits¹¹) cannot be tested with BRFSS. HCUP NEDS or SEDD data would be required.

No cost offsets. An earlier design envisioned SCM on T-MSIS/TAF Medicaid claims to estimate spending and service-mix effects. T-MSIS DUA access was not obtained for this draft.

5.3 Three paths for follow-up

Path 1: Extend BRFSS to 2024. BRFSS 2024 releases in mid-2025. Adding one post-treatment year moves the 2023 cohort (6 states: HI, KY, MD, MI, NH, TN) from not-yet-treated to post-treatment, increasing treated state-year observations from 9 to approximately 15. The 2024 cohort (5 states: CT, GA, MN, NE, NY) contributes first post-period observations. The additional year is likely sufficient to push marginal estimates across conventional thresholds if the underlying effects are real. Lowest-cost follow-up; should be the immediate next step.

Path 2: Medicaid claims validation via T-MSIS/TAF. The Transformed Medicaid Statistical Information System Analytic File provides state-year Medicaid claims including dental service utilization, enrollment, and spending. T-MSIS permits restriction to Medicaid-enrolled adults, eliminating state-population dilution, and supports total Medicaid spending (dental + medical + pharmacy) as an outcome. DUA process is typically 2–4 months and should run in parallel with Path 1.

Path 3: NHIS individual-level supplementary evidence. NHIS lacks state identifiers but has larger sample sizes and more detailed dental questions. A region-level supplementary analysis using individual-level NHIS data with a Medicaid-enrolled adult subsample would cross-check the BRFSS state estimates. NHIS DENTINT has gap years (2021, 2024) that modestly constrain the time dimension.

These three paths are complementary, not alternative. Together they would convert a one-dataset, underpowered BRFSS analysis into a three-dataset, adequately powered test of the cost offset and utilization hypotheses.

6. Conclusion

This paper asks whether the wave of twenty-six state Medicaid adult dental benefit expansions during 2020–2023 produced detectable changes in population-level dental utilization and access using BRFSS state-level data. The honest answer is: not in this data. Point estimates are directionally consistent with a modest positive effect on dental visits and a weak negative effect on cost-related care barriers, but confidence intervals include zero and the 3-period dental panel is structurally underpowered at the theoretically expected effect size.

This is a useful null in two senses. First, it establishes a concrete evidence baseline for state Medicaid directors considering similar expansions: short-term, state-population-level BRFSS dental visit changes in the first two years post-expansion should not be expected, because the dilution effect is strong and the BRFSS panel is thin. Second, it motivates the three follow-up paths that would convert this null into a credible detection: extending BRFSS to 2024, validating with T-MSIS/TAF claims, and supplementing with NHIS individual data.

The broader lesson is that rigorous state-level Medicaid policy evaluation requires data infrastructure that matches the granularity of the policy variation. BRFSS is the best available ongoing state-level survey and is not adequate for the question this paper attempts to answer. The future

of Medicaid policy evaluation lies in individual-level data with subgroup restriction, shorter-cadence state surveys, or — most promisingly — the T-MSIS/TAF claims infrastructure that CMS has been building over the past decade. Advancing each of these would enable the field to test the dental cost offset hypothesis with the precision it deserves.

Tables

Table 1. Primary outcome (past-year dental visit) difference-in-differences estimates. **Table 2.** Secondary outcome estimates (medcost, coverage, fair-poor health, diabetes) across 3-period and 8-year panels. **Table 3.** Romano-Wolf multiple-testing corrected p-values. See `analysis/tables/romano_wolf_corrected.csv`. **Table 4.** Pre-trend placebo regressions (8-year complete panel, 2016–2020). **Table 5.** NHIS Medicaid-enrolled dental visit trends. See `analysis/tables/medicaid_subsample_nhis.csv`. **Table 6.** Cohort-by-year grid for 19 treated states. See `analysis/tables/cohort_year_grid.csv`.

Figures

Figure 1. Raw past-year dental visit trends by treated / never-treated. **Figure 2.** Raw cost-related medical care barrier trends. **Figure 3.** Raw any-coverage trends. **Figure 4.** Raw fair-poor self-rated health trends.

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